

1. Question 1 The view of behavioral finance is that markets are always efficient and market participants are perfectly rational.
The correct answer is 'False'.
2. Question 2 Pareto efficiency is allocational efficiency, referring that in a financial equilibrium, allocations of assets are such that nobody's utility can decrease or increase by changing the allocations.
The correct answer is 'True'.
3. Question 3 The alpha is the difference between the mean of an asset and the risk-free return.
False
4. Question 4 Combining two risky assets can yield a portfolio with zero variance if the two assets are perfectly positively correlated (correlation coefficient is +1).
The correct answer is 'False'.
5. Question 5 If a utility function is strictly monotonic, then it is weakly monotonic as well.
The correct answer is 'True'
6. Question 6 One of the key ideas of Prospect Theory is that people frame their decisions in gains and losses rather than considering their potential final wealth.
The correct answer is 'True'.
7. Question 7 If markets are complete, then all second period consumptions can be achieved by asset trades.
The correct answer is 'True'.
8. Question 8 For weakly monotonic utility functions an arbitrage is a trading strategy that leads to strictly positive payoffs in all future states without requiring any payments.
The correct answer is 'True'.
9. Question 9 Risk neutral probabilities are representing the heterogeneous beliefs of the investors.
The correct answer is 'False'.
10. Question 10 The efficient frontier is the set of all portfolios in the mean-variance opportunity set for which there is no other point in the mean-variance opportunity set that improves mean and variance.
The correct answer is 'True'.

11. Question 11 Intertemporal trade on financial markets can be interpreted as follows: | need the money later, you need it now, let's trade!
The correct answer is 'True'.
12. Question 12 Mean and variance are not sufficient to describe highly skewed distributions, as they are typical for hedge funds or other alternative investments.
The correct answer is 'True'.
13. Question 13 Hyperbolic discounting is time-consistent, classical discounting is not.
The correct answer is 'False'.
14. Question 14 A limitation of CAPM is that it is not possible to study the effect of differences in the investors' beliefs.
The correct answer is 'False'.
15. Question 15 For weakly monotonic utility functions the absence of arbitrage is equivalent to the existence of non-negative state price with at least one strictly positive state price.
The correct answer is 'True'.
16. Question 16 There cannot be any arbitrage opportunities in the financial market equilibria.
The correct answer is 'True'.
17. Question 17 According to the Two-Fund Separation Theorem for mean-variance investors, all investors should invest to the market portfolio and to the risk free asset.
The correct answer is 'True'.
18. Question 18 If u is an expected utility function describing a person's preference relation $\prec$, then $v(x) = (u(x))^5$ is a utility function that describes $\prec$.
The correct answer is 'False'.

If u is an expected utility function describing a person's preference relation $\prec$, then $v(x) = (u(x))^5$ is a utility function that describes $\prec$.

Select one:

☐ True

☒ False ✓

19. Question 19 Agents in financial economics can trade consumption, opinions, financial assets, and risk factors.
The correct answer is 'True'.
20. Question 20 By definition, lottery A state dominates lottery B if A gives a strictly higher outcome than B in every state.
The correct answer is 'False'.
21. Question 21. What is the St. Petersburg Paradox? How can it be resolved?
"the player has to play infinitely many times. B" not play but pay a finite amount.

22. Question 22. List the names of the four axioms of expected utility theory?
- Completeness
 - Transitivity
 - Independence
 - Continuity
23. Question 23 Financial markets are useless, since the hunt for Alphas is a zero-sum game. Discuss.
- Investors only can win at the expense of someone else or the inefficiencies left behind by other less informed, intelligent or less savvy investors and it is consistent with the efficient market hypothesis. However investors can trade time risk and beliefs.
24. Question 24 How can one explain that Insurance Linked Securities (ILS) are not as popular as they should be according to CAPM with mean-variance investors?
- The return of distribution of ILS is very fat tailed to the left which means that there can be extreme outcomes (losses) and every now and then the real catastrophe happens and the investors have to face a huge loss.
25. Question 25 Consider a CAPM model with heterogenous investors, having mean-variance utility functions. What are active investors doing in this model? What are passive investors doing? How does being active or passive depend intuitively on a) market efficiency, b) the skill of the investor and c) the costs to be active?
26. Question 26 What is the law of one price when considering same payoffs generated by different combinations of assets? In practice, why is it possible that the law of one price does not hold? List at least two reasons in general or examples from the book in one sentence each.
27. It is possible that Law of one price does not hold because of the same reason as 3Com and Palm reasons that the companies and the investors wanted to create an arbitrage.

Question 27

Complete

Mark 3.50 out of 4.00

🚩 Flag question

Problem 27. (points: 4)

An investor's decisions can be explained by the expected utility function having $u(x) = -10e^{-2021x}$. How much is the absolute risk aversion of the investor? How does it depend on wealth (x)?

Problem 27

$$u(x) = -10e^{-2021x}$$

$$u' = -10(-2021)e^{-2021x}$$

$$u'' = -10(-2021^2)e^{-2021x}$$

$$r(x) = \frac{-10(-2021^2)e^{-2021x}}{-10(-2021)e^{-2021x}} = 2021$$

$R(x)$ should be proportional to x

Question 28

Complete

Mark 2.00 out of 2.00

Flag question

Problem 28. (points: 2)

On March 2, 2000, the company 3Com (having 47500 3Com stocks) made an IPO of one of its most profitable units. They decided to sell 5% of its Palm stocks and retain 95% thereof (95 000 Palm stocks). The Palm stock price closed at \$96.08. The mother company 3Com closed that day on \$80.19. How much is the value of Palm shares per 3Com share? Under what conditions could there be an arbitrage? List at least one.

Problem 28

3Com 47500

1 3Com is worth $95000/47500 = 2$ PALM shares

$2 \times 96.08 = 192.16$ value of Palm shares per 3Com

The arbitrage opportunity would be long 3Com & short Palm
However shorting is not possible FTA: depending on the utility function there is no arbitrage if and only if you have positive state price.

Problem 29. (points: 11)

An investor with mean-variance utility $U(\mu, \sigma) := \mu - \sigma^2$ can invest in two risky assets and one risk free asset. The risk-free return is 3%. The expected returns of the risky assets are $\mu_1 = 5\%$ and $\mu_2 = 7\%$. The covariance matrix is:

$$\begin{pmatrix} 2\% & -2\% \\ -2\% & 4\% \end{pmatrix}.$$

- Plot the indifference curves (σ along the horizontal axis and μ along the vertical axis) for utility levels of zero and one. (1 point)
- Calculate the optimal portfolio of the investor, including the weight of the risk free asset. (4 points)
- Calculate the investor's utility at the optimal portfolio. (2 points)
- Calculate the tangent portfolio. (1 point)
- Now consider a 3rd asset with mean $\mu_3 = 9\%$, extending the covariance matrix as follows.

$$\begin{pmatrix} 2\% & -2\% & -2\% \\ -2\% & 4\% & 5\% \\ -2\% & 5\% & 8\% \end{pmatrix}.$$

Calculate the beta and the alpha of the 3rd asset, with respect to the optimal portfolio of the first two assets (not the tangent portfolio). Interpret alpha. (3 points)

Problem 30. (points: 4)

Consider the following game: you flip a coin, if it is head, you win EUR 103 600, otherwise you win nothing. You can play only once. Let us assume your expected utility function is given by $u(x) := \sqrt{x}$ and your initial wealth is EUR 90 000.

- How big is your expected utility after playing this game?
- How much money is the certainty equivalent of the game? How is it related to the expected money you can win? Why?
- At most how much would you pay for the game (having the utility function specified above)?

Problem 31. (points: 4)

Consider an economy with two time periods, complete markets and uncertainty. Ignore consumption in the first period. There are two agents with exogenous income of $(w_1^1, w_1^2) = (1, 0)$ and $(w_2^1, w_2^2) = (0, 1)$ respectively in the two states of the second period. There are no endowments in the first period. Let π_s be the state price in state s . The agents' utility functions are given by:

$$U^1(c_1^1, c_2^1) = \ln(c_1^1) + \ln(c_2^1),$$

$$U^2(c_1^2, c_2^2) = \ln(c_1^2) + \ln(c_2^2).$$

Write the conditions for the competitive equilibrium. (You do not have to solve it). What is the motive for trade here?

Problem 31

Agent 1 $\max u^1(c_1^1, c_2^1) = \ln(c_1^1) + \ln(c_2^1)$

$$(w_1^1, w_2^1) = (1, 0)$$

such that

$$c_1^1 + \pi c_2^1 = w_1^1 + \pi w_2^1$$

Agent 2 $\max u^2(c_1^2, c_2^2) = \ln(c_1^2) + \ln(c_2^2)$

$$(w_1^2, w_2^2) = (0, 1)$$

such that

$$c_1^2 + \pi c_2^2 = w_1^2 + \pi w_2^2$$

Comment:
markets clear missing

Question 32

Complete

Mark 3.00 out of 3.00

Flag question

Problem 32. (points: 3)

Suppose we have financial markets with the following payoff matrix and market equilibrium prices

$$A = \begin{pmatrix} 1 & 4 \\ 1 & 3 \end{pmatrix} q = (1, 1).$$

Are markets arbitrage free? Argue using the (first) fundamental theorem of asset prices.

Prob. 32

~~Yes arbitrage free because the market is complete~~

Strictly positive π_1 & π_2 . If arbitrage free

$$\begin{cases} 1 \cdot \pi_1 + 1 \cdot \pi_2 = 1 \\ 4 \cdot \pi_1 + 3 \cdot \pi_2 = 1 \end{cases} \quad \pi_1 = 1 - \pi_2$$

$$4(1 - \pi_2) + 3\pi_2 = 1$$

$$4 - 4\pi_2 + 3\pi_2 = 1$$

$$-\pi_2 = -3$$

$$\pi_2 = 3 > 0$$

$$\pi_1 = 1 - 3 = -2 < 0$$

Not arbitrage free

Problem 33. (points: 7)

Suppose we have financial markets with the following payoff matrix and market equilibrium prices

$$A = \begin{pmatrix} 1 & 8 \\ 1 & 4 \\ 1 & 0 \end{pmatrix} \quad q = (1, 4).$$

- Are markets complete? Prove using the second fundamental theorem of asset prices by calculating state prices. (3 points)
- Introduce a new asset with a payoff $(0, 0, 3)$. This can be a put option written on the second asset with a strike price of 3. Find the price or pricing bounds using the no-arbitrage condition with non-negative state prices. (1 point)
- Consider a financial economy with a representative agent. Let the statistical probabilities (according to the belief of the representative agent) be equal. Let the representative agent's utility be $u(x) = \ln(x)$ (where x is consumption in any state) with a subjective discount factor of 1. Let the agent's endowment be $w_0 = 1, w_1 = 100/105, w_2 = 10/9, w_3 = 100/105$. Calculate the equilibrium state price in state 3 (π_3 only) and the equilibrium price of the new asset. (3 points)

33. MULTIPLE CHOICE QUESTIONS

1. It is a usual assumption in Finance that

- The price in the future follows normal distribution and the logreturn is lognormally Distributed.
- The price in the future follows normal distribution and the logreturn is normally Distributed.
- The price in the future follows lognormal distribution and the logreturn is log normally distributed.
- The price in the future follows lognormal distribution and the logreturn is Normally distributed.**

2. Value at risk is a given value of the

- a) Distribution function
- b) Cumulative distribution function
- c) **Inverse of the cumulative distribution function**
- d) Characteristic function

3. Which parameters of the weighting function are to be estimated by the financial Institution, if it applies the foundation internal ratings based approach for calculation of The capital requirement for credit risk?

- a) **PD**
- b) PD és LGD
- c) PD, LGD, EAD
- d) PD, LGD, EAD, M

4. What is the purpose of Business Impact Analysis?

- a) **Selection of critical business processes**
- b) Evaluation of crisis situation
- c) Informing senior management
- d) Communication

5. What does the external approach to crisis management focus on after a crisis situation?

- a) The learning process of the company
- b) **Social judgment**
- c) Leadership qualities
- d) Building external relationships

TRUE/FALSE

1. Value at Risk is a coherent risk measure. (False)
2. The VaR of portfolio equals to the sum of the VaR measures of the subportfolios only if the subportfolios are perfectly correlated (correlation coefficient is 1). (True)
3. Harris Poll Reputation Index includes the social responsibility. (True)
4. The stock price of a company is suitable for measuring reputation risk. (True)
5. The legal risk is not operational risk. (False)
6. The volatility of return is proportional to the time (False)
7. Delta normal approach assumes the normal distribution of the risk factor (True)
8. Value at -risk has two parameters: time and significance level (True)
9. ERM concept is a top-down approach. (True)
10. Conditional default probabilities are always less than the unconditional ones. (False)

11. What is the focus of the internal approach?
Preparedness of the organization.

12. Which is not true for basel 2?

Capital requirements are calculated for markets credits and operational risks

1. **What is in the focus of the internal approach to crisis management?**
 - a. **Preparedness of the organization**

0.

1. What is **not** true for BASEL II

- a. **It extends the regulation by different capital buffer requirements**

Logic: BUFFER IS FOR BASEL III

3 pillars exist (2007)

- CapReq (credit Op. Risk)
- Supervisory Review
- Market Discipline

0. In Monte Carlo Simulation we can assume non-normal distributions for the random variable.

- **True**

Logic: MC can use data with a high standard deviation and can assume distribution other than the normal.

0. In case of an internal fraud the defensive crisis reporting strategy is appropriate.

- False**

Logic: Defensive strategy → External approach during crisis

0. The VaR of a portfolio equals to the sum of the VaR measures of the subportfolios only if the correlation coefficient between the subportfolios equals to

- a. **1**

Logic: Perfect positive correlation i.e. 1.

0. The role of scenario analysis is to evaluate the low impact – high probability risks.

- **False**

Logic: THE OTHER WAY AROUND scenario analysis focuses on rare (low probability events with high impact)

0. Which metric is not included in measuring reputation risk?

- Receivables**

0. Regulators are planning to move toward a system where ES (Expected Shortfall) is used as measure.

- **True**

0. Cash-flow at risk is a risk measure often used in corporate risk-management.

- **True**

0. How would you model the loss deriving from internal fraud under the advanced measurement approach?

0. What are the advantages and disadvantages of the VaR?

Advantages (slide 24):

- It captures an important aspect of risk in a single number.
- It is easy to understand.
- It asks the simple question: “How bad can things get?”

Disadvantages (google)

- difficult to calculate for large portfolios
- VaR is not additive , $VaR(A+B) \neq VaR(A) + VaR(B)$
- Different methods to calculate VaR give different results.

0. What are the risk factors affecting the bond portfolio? How would you limit the potential loss of your portfolio?

risk factors :

- interest rate risk.
- credit risk
- default risk
- years of compounded growth (compounded or continuous).

limit potential losses by diversification.

0. A portfolio of two risky assets has the lowest risk measured by the variance if the correlation coefficient of their returns is

-1

0. Which event may pose the most significant reputational risk?

a. Internal fraud of the management

Which statement is false? Normal distribution

a. Has fat-tails

0. What is in the focus of the internal approach to crisis management?

a. For the preparedness of the organization

0. Stress testing involves evaluating the impact of extreme, but plausible, scenarios.

- **True**

Logic: Stress testing focuses on 1 event /activity under extreme circumstances.

0. Capital requirement of credit risk is 10% of the value of the risk-weighted assets.

- **False**

Logic: 1988: BIS Accord Capital must be **8%** of risk weighted amount.

0. Which concept is among the operational responses?

a. Protection of human life

0. What does the external approach to crisis management focus on after a crisis situation?

a. For social judgment of the company

0. The relationship between the value of a bond and the level of the yield curve is linear.

- **False**

Logic: convex

0. Default occurs in the Merton Model if the value of the firm's assets is below the notional of the bonds at maturity.

- **True**

0. Which risk was not included in the top 10 operational risks in 2020?

a. **War**

0. Which is **not** true for reputation risk?

a. **Easy to measure and quantify**

0. Which factor is **not** included in the Harris Poll Reputation Index?

a. **Age of the customers**

0. The AMA (Advanced Measurement Approach) method used to calculate the operational risk capital requirement is **not** subject to a supervisory authorization.

- **False**

0. **Suppose that a 4-year zero-coupon Treasury bond with a face value of 100 yields 3% and a similar 4-year zero-coupon bond issued by a corporation yields 4%. (Both rates are logreturns.) Make the simplifying assumption that there are no recoveries in the event of a default. Calculate the default probability of the corporation!**

a. **3.92%**

0. Which parameter (s) of the weighting function is/are to be estimated by the financial institution if it applies the foundation internal ratings-based approach for the calculation of the capital requirement for credit risk?

a. **PD**

0. Which statement is true?

a. **If two random variables are independent, their correlation is zero.**

0. A transition matrix contains the probabilities of changing the rating within a given period.

- **True**

0. What is the expected default frequency if the distance to default is zero?

- a. 50%
0. Corporations do not need to manage their risks, as they are not regulated.
- False
0. Using historical simulation, we get always a higher risk measure than using the delta-normal method.
- False
0. The advantage of the ratings given by rating agencies is that they react fast to the new information.
- False
0. The maximal loss of a long option position is the value of the option.
- True
0. Value at risk is a given value of the:
- a. Inverse of the cumulative distribution function
0. Regulators are planning to move toward a system where ES is used as a risk measure.
- True
0. What is the purpose of the Business Impact Analysis?
- a. Selection of critical business processes
0. The 1-day Value at Risk of a portfolio at 95% significance level is 5 (million) USD. How much is the 10-day VaR at 95% significance of the same portfolio, if the portfolio's return is supposed to follow a normal distribution with zero mean.
- a. 15.81 MUSD
0. Which function shows the probability that the value of a certain random variable won't exceed?
- a. Cumulative distribution function

The role of scenario analysis is to evaluate the low impact – high probability risks.

- False

43. what is the correct comparison between the value of European American and Bermudan put option

European < Bermudan < American

a. 6.7

Delta normal		99%	95%
	mean	\$0.00	\$0.00
	stdev	\$1.29	\$4.08
	days	1	10
	multiplier	2.33	1.64
	VaR	\$3.00	\$6.71

45. what could be the best definition of DSCR?

b. The free cash flow of a period over the debt service of the period

46. Consider a portfolio of two securities one share of YYY stock and a bond that pays 100 in one year. Suppose this portfolio is currently trading with a bid price of 171.22 and an ask price of 174.11 and the bond price of 95.5.

c. 73.32-78.61

Arbitrage: buy (ask) cheap, sell (bid) at a higher "expensive" price. NO arbitrage means this possibility is eliminated.									
			Bid	Ask					
1 YYY + 1 Bond			171.22	174.11	you can buy the portfolio at 174.11 (or sell it at 171.22)				
	Bond		95.5	97.9	and sell the bond at 95.5 (or buy the bond at 97.9)				
	YYY?		73.32	78.61	you are left with a stock worth 174.11 - 95.5 = 78.61 as if you bought it directly at Ask (you are left with cash of 171.22 - 97.9 = 73.32 as if you originally sold the stock at Bid)				

47. Which of the following statement is true?

C. Prospect theory violates stochastic dominance in an example where several small probability events with similar outcome are outweighed relative to a single outcome with a slight larger payoff.

48. Read all the possible answers below.

A leasing has to be treated as a finance lease if...

c the lease contains an option to purchase the asset at a bargain price that is substantially less than its fair market value.

49. According to prospect theory, people gamble and buy insurance because of people overweight several small probability events with similar outcome are overweighted relative to a single outcome with a slight larger payoff. TRUE

50. Which statement is true?

a. The average of the result from different valuation approach does not lead to meaningful result

yes, it DOES NOT because assumptions for each approach are different

51. Assume WACC=10% rD=5% t=20% D/V=11% RONIC=12% g=6% NOPLAT=200 Which statement is true?

c. Only that the operational cost of capital is 10.4%

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

cost of debt ==> rD	
K is only relevant for the TV period and since WACC < ROIC < RONIC, TV is not applicable so no K is applicable	
	0.50
rA 0.4*5%+0.6*0.15	0.10 4
rE = (WACC - D/V * rD * (1 - t)) (1 - D/V) (10%-0.4*5%*0.8) / 0.6	0.14 0

52. Which of the following statement is false

a. if the equilibrium prices of all assets are unique, then markets are complete

if the eq. prices of ALL ASSETS are ARBITRAGE FREE then markets are complete

53. Which of the following positions is not sensitive on volatility?

a. Long forward

Long forward position (and any position in a forward or in the stock itself) does not depend on volatility.

54. Which statement is true?

c. Multiple base valuation methods assume that the firm has the same business outlook as its peers.

The process of multiple-based valuation

0. Pick a multiple.

1. Choose comparable firms. Similar risk, growth, profit generation
(No industry or size requirement theoretically.)

2. Control for differences. (Similar but not the same.)

Subjective corrections (weighted average)

Modified multiples (corrected for differences)

3. Check reasonability.

Industry regressions

Market regressions

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Principles of Corporate Finance, 6th edition

Equity has to be **less than or equal** to Asset

The TV formula for **FCFF, APV, and FCFE** can be interpreted as a growing perp.

The **Multiple-based methods** show the likely price a third-party buyer may pay for the given firm

55. which of the following statement is false?

c. since markets are complete, in Bloomberg you can choose like 5 models to price complicated options each giving a different price
Since markets are complete, (..), each giving very **similar prices**

56. What is the position of the bondholder in the merton model?

b. Short put

57. What does FRA mean?

b. Forward rate agreement

58. Which of the following is not in an operational risk.

d. Raising the central bank base rate.

59. Which type of ratios use data from the income statement and the balance sheet as well?

a. working capital turnover ratios

60. which parameter of the weighting function is are to be estimated by the financial institution if iit applies the foundation internal ratings based approach for the calculation of the capital requirement for credit risk?

PD.

61. you are an owner of a firm in which firm do you have a limited liability?

d. corporation

62. what is the typical for project loan financing?

a. contract based financing

63. the zero coupon rate curve (continuously compounding) is flat at 5%. what is the fair price of a 3 year zero coupon bond?

a. 86.07%

$$100 * \text{EXP}(-5\% * 3)$$

Duration	3	Time of Maturity has the only payment
YTM	5%	because it is the only rate of discount and it's a flat curve

64. Which one of the following is not a standard covenant?

d. no dividend payment is allowed.

Standard covenants

- Cross default
- Ownership clause
- Pari passu
- Negative pledge
- Fresh borrowing
- Information provision

65. the market price of down and knock out call option is 5. the price of a vanilla call with the same parameter is 7. what is the fair price for the down and in knock in call option with the same parameters?

c. 2

Vanilla = KnockIN + KnockOUT

7 = 2 + 5

66. Which statement is true?

a. The criterion of uniformity states that we have to value all assets using the same methodology.

relevance applies for **selecting the right multiple**

equals **WACC**

uniformity applies **to selecting the right multiple for calculating the value of the firm**

67. How should we hedge a short stock call position?

a. Long stock

68. The Weiner process?

a. is a martingale and it is markovian.

Present value approach:

$$g = (\text{Noplat}_t / \text{Noplat}_{t-1}) - 1$$

$$\text{Noplat}_t = \text{EBIT}_t * (1 - \text{Tax rate})$$

$$\text{Noplat}_t = \Delta \text{invested Capital}_t / K_t$$

$$K_t = \Delta \text{Invested Capital}_t / \text{Noplat}_t$$

$$K_t = g_{t+1} / \text{RONIC}_{t+1}$$

$$\text{ROIC}_t = \text{Noplat}_t / \text{Invested Capital}_{t-1}$$

$$\text{RONIC}_t = \Delta \text{Noplat}_t / \Delta \text{Invested Capital}_{t-1}$$

$$FCFF_t = NOPLAT_t - \Delta \text{Invested Capital}_t$$

$$FCFF_t = NOPLAT_t - NOPLAT_t * K_t$$

$$FCFF_t = NOPLAT_t * (1 - K_t)$$

$$FCFF = FCFD + FCFE$$

$$FCFD = \Delta \text{Debt}_t - \text{Interest} + \text{Tax shield (TS)}$$

$$FCFE = \Delta \text{Equity}_t (\text{Required Equity}) - \text{Retained Earnings} - \text{Dividends}$$

$$\text{EVA} = \text{Actual Profit from operations} - \text{Required Profit from Invested Capital}$$

$$= NOPLAT_t - (IC_{t-1} * WACC_t)$$

$$= IC_{t-1} * (ROIC_t - WACC_t)$$

$$RI = \text{Actual Profit for the year for equity holders} - \text{Required Profit by equity Holders}$$

$$= PAT - (BV.Equity_{t-1} * rE)$$

$$= BV.Equity_{t-1} * (ROE_t - rE_t)$$

$$rA = D/V * rD + E/V * rE$$

$$WACC = D/V * rD * (1 - \text{Tax rate}) + E/V * rE$$

$$WACC = rA - D/V * rD * t$$

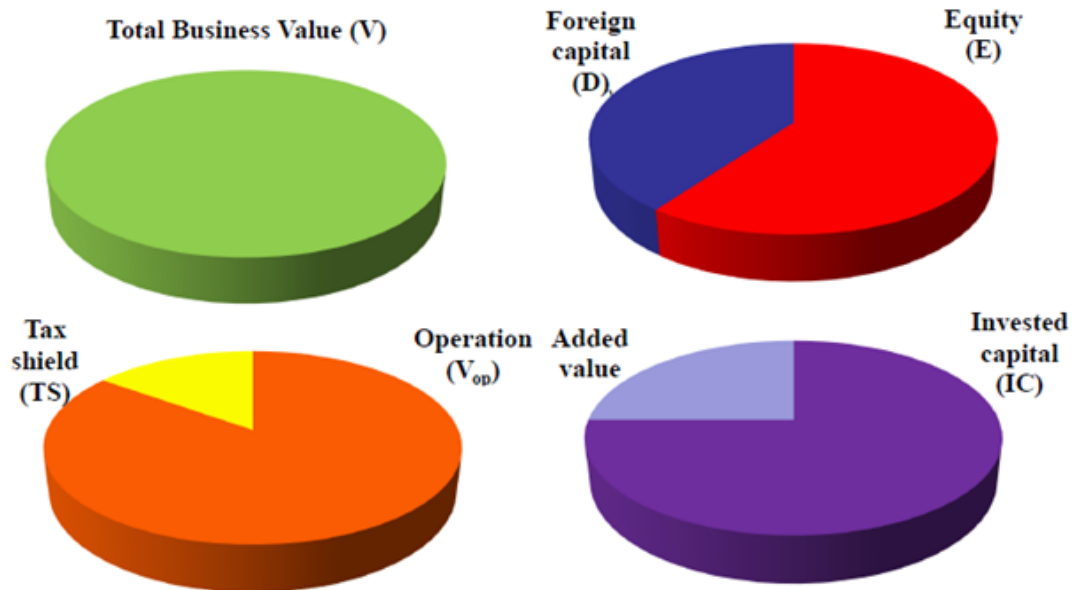
$$rE = (rA - D/V * rD) / (1 - D/V)$$

DCF Models

Entity Model (V – D = E)	Equity Model (E)
FCFF	FCFE
APV	RI
EVA	

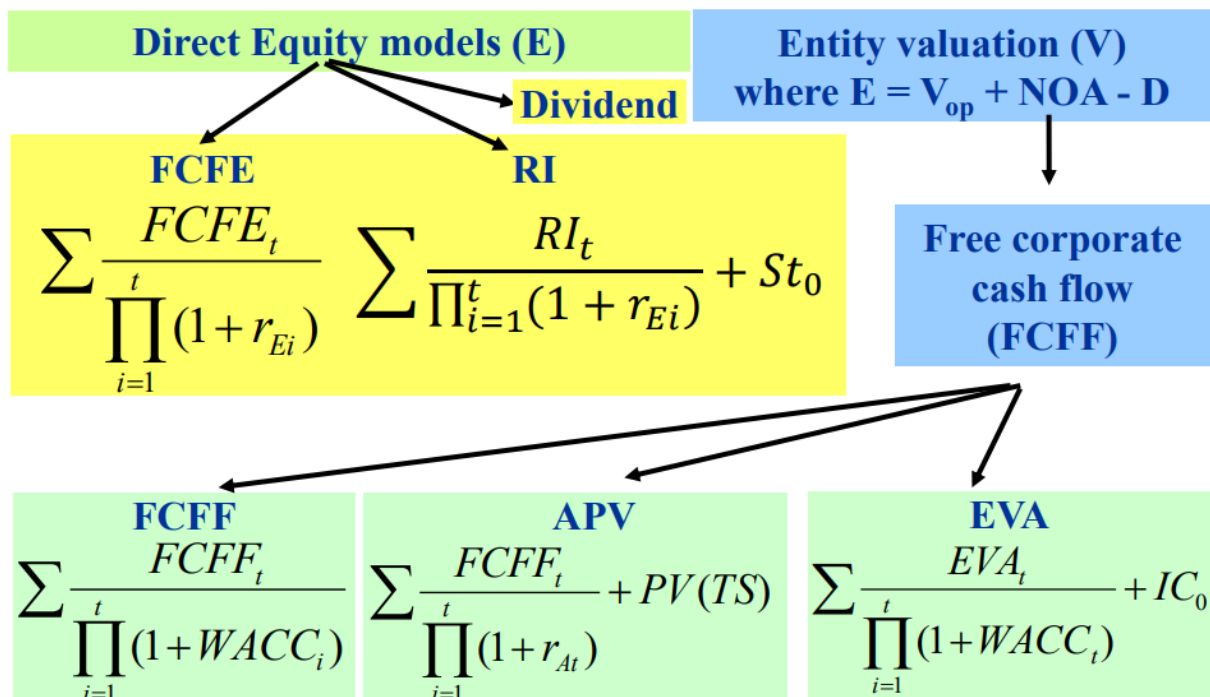
Model	Cost of Capital/Discount rate	Formula	Terminal Period
FCFF	WACC	$SUM(FCFF/(1+WACC))$	$FCFF/(WACC-g)$
APV	rA	$SUM((FCFF+TS)/(1+rA))$	$(FCFF+TS)/(rA-g)$
FCFE	rE	$SUM(FCFE/(1+rE))$	$FCFE/(rE-g)$
EVA	WACC	$SUM(EVA/(1+WACC)) + IC_{boy} = V_{boy}$	$V_{boy} - IC_{boy} = NOPLAT/WACC$
RI	rE	$SUM(RI/(1+rE)) + BV.Equity = E_{boy}$	$E_{boy} - BV.Equity = PAT/rE$

The division of the business value



17/41

Method	To discount	Discount rate	Notes
Total firm DCF	FCFF	WACC	Useful for multiple business units and set D/V ratio firms
Economic profit	EVA / RI	WACC / r_E	Shows when the value will be created.
APV	FCFF and tax shield (TS)	Operative cost of capital (r_A)	Separates the effect of financing, handy for M&As.
Direct equity valuation	FCFE	Required rate on equity (r_E)	Detailed, hectic financial plan available, bottom-up CF, financial service industry



Why do we have various DCF methods? When is it most handy to use one or the other?

The reason we have various DCF methods - which should all theoretically result in the exact same value based on MM theorems - is because the total business value can be dissected in more than one way.

The Total firm DCF method is used when multiple business units should be valued and when D/V ratio could be assumed, as it is essential for WACC calculation (discount rate used for this method).

Based on the Direct Equity Valuation model, Equity and Foreign capital (i.e. Debt) are the two components that make up the total firm value. This model is useful when a detailed financial plan is available or when a bottom-up CF approach should be followed. It is also widely used in the financial service industry.

APV separates the effect of financing by assuming the total value to be composed of the Operational value (FCFF) and the Tax shield (TS). This method is handy for M&As.

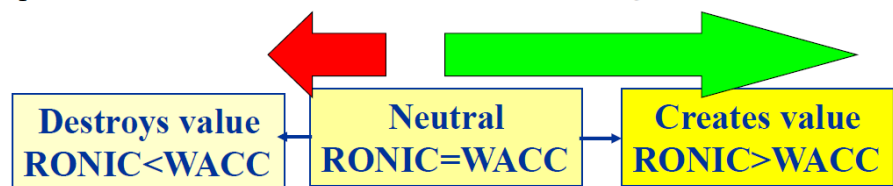
Economic profit models such as EVA and RI show when the value will be created.

g, ROIC, WACC and the business value

		RONIC				
		6%	8%	10%	12%	14%
g	4%	709	891	1 000	1 073	1 125
	6%	495	811	1 000	1 126	1 217
	8%	215	706	1 000	1 196	1 336

$IC_0=1000$, $ROIC_1=RONIC$, $WACC=10\%$, after 20 years $RONIC=WACC$

Growth:



Bond pricing:

Zero Rate/Zero-Coupon Rate	
Duration	T
$YTM = -\ln(\text{Discount factor})/T$	
Bond Price and Duration are inversely related	
Modified Duration = $D/(1+y/m)$	

Option Pricing

This topic gives us a clear understanding of how the different options are valued, ie, Vanilla and Exotic options.

Types of Options	Analytic	Binomial Tree	Monte Carlo
European Vanilla	yes BS formula	YES	YES
American	for call also BS	YES	NO
Asian	NO	YES, BUT only nonrecombining	YES
Bermuda	not sure about this one	YES	YES
Barrier	YES	YES	YES, Monte Carlo is good for Barrier Options

Types of options

Bermudan option

A Bermudan option is an option that gives the holder the right to exercise a set number of times. These options fall between European and American options.

Bermudan options allow investors to buy or sell an underlying asset or security at a pre-set price. However, it does not specify a single date or allow exercise at any time. Instead, it sets several dates along the expiration date to exercise.

Asian option

Asian options are options where the payoff depends on the arithmetic average of the price of the underlying asset during the life of the option.

We have two types of Asian Options

Fixed Strike

Here, the strike price is constant while the stock average price varies. See in the excel file

Fixed Price

We use the Stock price at maturity and stock average price. We do not use strike price here. See in the excel file.

American option

American options can be exercised at any time up to the expiration date.

European Option

European options can be exercised only on the expiration date itself.

European options are generally easier to analyse than American options.

Pricing methods

1. Binomial pricing

See attached Binomial pricing File

o. Monte Carlo Pricing

Monte Carlo simulation involves using random numbers to sample many different paths that the variables underlying the derivative could follow in a risk-neutral world. For each path, the payoff is calculated and discounted at the risk-free interest rate. The arithmetic average of the discounted payoffs is the estimated value of the derivative.

See attached Monte Carlo pricing File

o. Black Scholes Pricing

Using the Black Scholes differential equation, we can value our option. The differential equation for the price of any derivative dependent on a stock can be obtained by creating a riskless portfolio of the derivative and the stock. Because the derivative's price and the stock price both depend on the same underlying source of uncertainty, this can always be done. The portfolio that is created remains riskless for only a very short period of time. However, the return on a riskless portfolio must always be the risk-free interest rate if there are to be no arbitrage opportunities. The expected return on the stock does not enter into the Black–Scholes–Merton differential equation. This leads to an extremely useful result known as risk-neutral valuation. This result states that when valuing a derivative dependent on a stock price, we can assume that the world is risk neutral. This means that we can assume that the expected return from the stock is the risk-free interest rate, and then discount expected payoffs at the risk-free interest rate. The Black–Scholes–Merton equations for European call and put options can be derived by either solving their differential equation or by using risk-neutral valuation.

See Balsam File on Black Scholes.

It is worth remembering the concept of Put-call parity.

CHAPTER 10

Properties of Stock Options

Practice Questions

Problem 10.8.

Explain why the arguments leading to put–call parity for European options cannot be used to give a similar result for American options.

When early exercise is not possible, we can argue that two portfolios that are worth the

same at time T must be worth the same at earlier times. When early exercise is possible, the argument falls down. Suppose that $P + S > C + Ke^{-rT}$. This situation does not lead to an arbitrage opportunity. If we buy the call, short the put, and short the stock, we cannot be sure of the result because we do not know when the put will be exercised.

Problem 10.9.

What is a lower bound for the price of a six-month call option on a non-dividend-paying stock when the stock price is \$80, the strike price is \$75, and the risk-free interest rate is 10% per annum?

The lower bound is

$$80 - 75e^{-0.1 \times 0.5} = \$8.66$$

Problem 10.10

What is a lower bound for the price of a two-month European put option on a non-dividend-paying stock when the stock price is \$58, the strike price is \$65, and the risk-free interest rate is 5% per annum?

The lower bound is

$$65e^{-0.05 \times 2/12} - 58 = \$6.46$$

Problem 10.13.

Give an intuitive explanation of why the early exercise of an American put becomes more attractive as the risk-free rate increases and volatility decreases.

The early exercise of an American put is attractive when the interest earned on the strike price is greater than the insurance element lost. When interest rates increase, the value of the interest earned on the strike price increases making early exercise more attractive. When volatility decreases, the insurance element is less valuable. Again, this makes early exercise more attractive.

Problem 10.14.

The price of a European call that expires in six months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in two months and again in five months. Risk-free interest rates for all maturities are 10%. What is the price of a European put option that expires in six months and has a strike price of \$30?

Using the notation in the chapter, put-call parity, equation (10.10), gives

$$c + Ke^{-rT} + D = p + S_0$$

or

$$p = c + Ke^{-rT} + D - S_0$$

In this case

$$p = 2 + 30e^{-0.1 \times 6/12} + (0.5e^{-0.1 \times 2/12} + 0.5e^{-0.1 \times 5/12}) - 29 = 2.51$$

In other words the put price is \$2.51.

1. Net working capital = current assets – current liabilities Important: cash that is not needed for the everyday operation is not part of the NWC.

2. Market value vs book value of equity

□ difference is because book value shows the historical costs; valuable assets are not captured (eg. quality of management team)

3. Market value of equity (E) =

= market capitalization = shares outstanding x market price per share □ value depends on what investors expect in the future.

4. Market to book ratio = market value of equity / book value of equity = $MV / BV = E / BV$

OR

price to book ratio = P / B (in this case we measure it on a single share's level) □ low: value stock

5. Enterprise value (EV) = Total enterprise value (TEV) □ operational value of the company = Market value of equity + Debt – cash (or any Non-Operating Asset (NOA) eg. marketable securities)

If Global has 3.6 million shares outstanding, and these shares are trading for a price of \$14 per share, what is Global's market capitalization compare to Global's book value of equity in 2015?

Global's market cap:

3.6 million shares x 14 \$/share = 50.4 million \$

Market to book ratio = $50.4 / 22.2 = 2.27$

□ this means that an investor is willing to pay 2.27 times more money for Global's share than its BV.

EPS = Net income / shares outstanding

Diluted EPS □ calculates

also with employers' and executives' stock options, and convertible bonds

Statement of Cash Flows I. / a

Utilizes the information from the BS and IS to determine how much cash the firm has generated, and how the cash has been allocated during a set period.

Adjusts net income by all non-cash items related to operating activities and changes in net working capital

- Depreciation – add the amount of depreciation

Operating activity Investment activity Financing activity

Capital Expenditures (CAPEX)

- Accounts Receivable – deduct the increases - Accounts Payable – add the increases -

Inventories – deduct the increases

Payment of dividends

Sale/purchase of stocks

Changes in borrowings

Buying or selling marketable securities

Statement of Cash Flows I. / b

$FCFF = FCFD + FCFE$

$FCFF = \text{Free Cash Flow to Firm} = EBIT \times (1 - \text{tax rate}) - \text{change in net working capital} - \text{change in long-term assets}$

FCFF definition: CF generated through the operation of the company, which can be freely paid out to equity-, and debtholders.

$FCFE = \text{Free Cash Flow to Equity} = \text{dividend payment} + \text{stock repurchase} - \text{stock sales}$

$FCFD = \text{change in debt} + \text{interest payment}$

Suppose global had an additional \$1 million depreciation expense in 2015. if Global's tax rate on pretax income is 26%, what would be the impact of this expenses on Global's earnings? How would it impact Global's cash balance at the end of the year?

Depreciation is an operation expense, so Global's operating income, EBIT, and pretax income would fall by \$1 million. This decrease in pretax income would reduce Global's tax bill by $26\% \times \$1 \text{ million} = \0.26 million . Therefore, net income would fall by $1 - 0.26 = \$0.74 \text{ million}$.

On the statement of cash flows, net income would fall by \$0.74 million, but we would add back the additional depreciation of \$1 million because it is not a cash expense. Thus cash from operating activities would rise by $-0.74 + 1 = \$0.26 \text{ million}$. Thus, global's cash balance at the end of the year would increase by \$0.26 million, the amount of the tax savings that resulted from the additional depreciation expense.

Statement of Stockholders' Equity & Other

Breaks down the stockholders' equity computed on the BS into the amount that came from issuing shares versus retained earnings.

$\text{Change in shareholders' equity} = \text{retained earnings} + \text{Net sales of stocks} = \text{Net income} - \text{dividends} + \text{sales of stocks} - \text{repurchase of stocks}$

Other:

- Management discussion and analysis (MD&A): Providing background to the company and any significant events that occurred. Discuss the future plans, goals, new projects, etc. + liquidity issues + off-balance sheet transactions
- Notes to the financial statements: further details, information related to the statements. Eg. Information on debts, leases, stock-based compensation plans...

Financial Statement Analysis I.

1. Firm with itself → firm has changed over time

2. Firm to other similar firms using a common set of financial ratios

3. Type of ratios:

- Profitability ratios → IS
- Liquidity ratios → BS
- Working capital ratios → IS & BS
- Interest coverage ratios → IS
- Leverage ratios
- Valuation ratios
- Operating returns

Profitability Ratios		Leverage Ratios (continued)	
Gross Margin [28%, 42%, 65%]	$\frac{\text{Gross Profit}}{\text{Sales}}$	Debt-to-Capital Ratio [18%, 38%, 56%]	$\frac{\text{Total Debt}}{\text{Total Equity} + \text{Total Debt}}$
Operating Margin [7%, 13%, 22%]	$\frac{\text{Operating Income}}{\text{Sales}}$	Debt-to-Enterprise Value Ratio [-4%, 9%, 25%]	$\frac{\text{Net Debt}}{\text{Enterprise Value}}$
EBIT Margin [6%, 12%, 20%]	$\frac{\text{EBIT}}{\text{Sales}}$	Equity Multiplier (book) [1.7x, 2.5x, 4.0x]	$\frac{\text{Total Assets}}{\text{Book Value of Equity}}$
Net Profit Margin [2%, 7%, 14%]	$\frac{\text{Net Income}}{\text{Sales}}$	Equity Multiplier (market) [1.0x, 1.1x, 1.5x]	$\frac{\text{Enterprise Value}}{\text{Market Value of Equity}}$
Liquidity Ratios		Valuation Ratios	
Current Ratio [1.2x, 1.8x, 2.9x]	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$	Market-to-Book Ratio [1.6x, 2.9x, 5.5x]	$\frac{\text{Market Value of Equity}}{\text{Book Value of Equity}}$
Quick Ratio [0.7x, 1.2x, 2.0x]	$\frac{\text{Cash \& Short-term Investments} + \text{Accounts Receivable}}{\text{Current Liabilities}}$	Price-Earnings Ratio [15.7x, 21.6x, 32.6x]	$\frac{\text{Share Price}}{\text{Earnings per Share}}$
Cash Ratio [0.1x, 0.4x, 0.8x]	$\frac{\text{Cash}}{\text{Current Liabilities}}$	Enterprise Value to Sales [1.3x, 2.4x, 4.3x]	$\frac{\text{Enterprise Value}}{\text{Sales}}$
Working Capital Ratios		Enterprise Value to EBIT [11.9x, 15.7x, 22.2x]	$\frac{\text{Enterprise Value}}{\text{EBIT}}$
Accounts Receivable Days [32, 49, 67]	$\frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$	Enterprise Value to EBITDA [8.8x, 11.5x, 15.4x]	$\frac{\text{Enterprise Value}}{\text{EBITDA}}$
Accounts Payable Days [25, 42, 62]	$\frac{\text{Accounts Payable}}{\text{Average Daily Cost of Sales}}$	Operating Returns	
Inventory Days [24, 54, 92]	$\frac{\text{Inventory}}{\text{Average Daily Cost of Sales}}$	Asset Turnover [0.3x, 0.6x, 1.1x]	$\frac{\text{Sales}}{\text{Total Assets}}$
Interest Coverage Ratios		Return on Equity (ROE) [4%, 11%, 19%]	$\frac{\text{Net Income}}{\text{Book Value of Equity}}$
EBIT/Interest Coverage [2.9x, 6.7x, 15.8x]	$\frac{\text{EBIT}}{\text{Interest Expense}}$	Return on Assets (ROA) [-1%, 3%, 8%]	$\frac{\text{Net Income} + \text{Interest Expense}}{\text{Book Value of Assets}}$
EBITDA/Interest Coverage [5.2x, 9.8x, 20.2x]	$\frac{\text{EBITDA}}{\text{Interest Expense}}$	Return on Invested Capital (ROIC) [7%, 12%, 21%]	$\frac{\text{EBIT} (1 - \text{Tax Rate})}{\text{Book Value of Equity} + \text{Net Debt}}$
Leverage Ratios			
Debt-Equity Ratio (book) [21%, 60%, 121%]	$\frac{\text{Total Debt}}{\text{Book Value of Equity}}$		
Debt-Equity Ratio (market) [6%, 21%, 51%]	$\frac{\text{Total Debt}}{\text{Market Value of Equity}}$		

Gross margin = gross profit / sales → shows the ability to sell a product for more than the cost of production

Operating margin = operating income / sales → efficiency of operation EBIT margin = EBIT / sales → efficiency of operation; different corporate strategy

Net profit margin = net income / sales → shows the fraction of each dollar in revenues that is available for equity holders (difference can come from efficiency OR leverage, OR accounting assumptions)

Financial Statement Analysis III.

Liquidity ratios

The firm has sufficient working capital to meet its short-term needs.

Current ratio = current assets / current liabilities

Quick ratio = (cash + short term inv. + acc.receivable) / current liabilities
Cash ratio = cash / current liabilities

Financial Statement Analysis IV.

Working capital ratios

How efficiently the firm is utilizing its net working capital.

AccRec. days = $AR * 365 / \text{sales}$ the speed at which a company turns sales into cash

AccPay. days = $AP * 365 / \text{cost of sales}$

Inventory days = $\text{inventory} * 365 / (\text{cost of sales} + \text{SG\&A costs})$

AR turnover = Sales / AR

AP turnover = $\text{Cost of sales} / AP$

Inventory turnover = $(\text{Cost of sales} + \text{SG\&A costs}) / \text{inventory}$

Financial Statement Analysis V.

Interest coverage ratios

EBIT / interest

EBITDA / interest

(EBITDA = EBIT + depreciation and amortization) 22

Financial Statement Analysis V.

Leverage ratios

The extent to which the company relies on debt as a source of financing.

Debt-Equity ratio = $\text{Total Debt} / \text{Total Equity}$

Debt-to-Capital ratio = $\text{Total Debt} / (\text{Total Equity} + \text{Total Debt})$
Net Debt = $\text{Total Debt} - \text{Excess cash and short-term investment}$
Debt-to-Enterprise value = $\text{Net Debt} / (\text{Market Value of Equity} + \text{Net Debt})$

Equity multiplier = $\text{Total Assets} / \text{Book value of Equity}$

Financial Statement Analysis VI.

Valuation ratios

Price-earnings (P/E) = $\text{Market capitalization}_{t-1} / \text{Net income}_t = \text{Share price}_{t-1} / \text{EPS}_{t-1}$!!!!

Sensitive to the choice of leverage, and depreciation and amortization.

EV / Sales

EV / EBITDA

EV/EBIT

24

Financial Statement Analysis VII.

Operation returns

Return on Equity (ROE) = $\text{Net Income}_t / \text{Book Value of Equity}_{t-1}$

ROE provides a measure of the return that the firm has earned on its past investments.

Return on Assets (ROA) = $(\text{Net Income} + \text{Interest expense})_t / \text{Book Value of Assets}_{t-1}$

Return on Invested Capital (ROIC) = $\text{EBIT} \times (1 - \text{tax rate})_t / (\text{Book Value of Equity}_{t-1} + \text{Net Debt}_{t-1})$

ROIC is the most useful in assessing the performance of the underlying business.

EBIT x (1-tax rate) = NOPLAT = Net Operating Profit Less Adjusted Taxes

Financial Statement Analysis VIII.

DuPont Identity

$ROE = \text{Net Income} / \text{Sales} \times \text{Sales} / \text{Total Assets} \times \text{Total Assets} / \text{Book Value of Equity}$
profit margin Asset turnover Equity Multiplier

According to the Two-Fund Separation Theorem for mean-variance investors, all investors should invest to the market portfolio and to the risk free asset.

The correct answer is 'True'.

According to the Two-Fund Separation Theorem for mean-variance investors, all investors should invest to the market portfolio and to the risk free asset

The correct answer is 'True'.

Risk neutral probabilities are representing the heterogeneous beliefs of the investors.

The correct answer is 'False'.

For weakly monotonic utility functions an arbitrage is a trading strategy that leads to strictly positive payoffs in all future states without requiring any payments.

The correct answer is 'True'.

A limitation of CAPM is that it is not possible to study the effect of differences in the investors' beliefs.

The correct answer is 'False'

Intertemporal trade on financial markets can be interpreted as follows: I need the money later, you need it now, let's trade!

The correct answer is 'True'.

There cannot be any arbitrage opportunities in the financial market equilibria.

The correct answer is 'True'.

List the names of the four axioms of expected utility theory.

- Completeness - Independence - Continuity – Transitivity

One of the key ideas of Prospect Theory is that people frame their decisions in gains and losses rather than considering their potential final wealth.

The correct answer is 'True'.

if u is an expected utility function describing a person's preference relation $\prec$ then

$u(x) = (u(x))^5$ is a utility function that describes $\prec$.

The correct answer is 'False'

Mean and variance are not sufficient to describe highly skewed distributions, as they are typical for hedge funds or other alternative investments.

The correct answer is 'True'.

The view of behavioral finance is that markets are always efficient and market participants are perfectly rational.
The correct answer is 'False'.

For weakly monotonic utility functions the absence of arbitrage is equivalent to the existence of non-negative state price with at least one strictly positive state price.
The correct answer is 'True'.

Agents in financial economics can trade consumption, opinions, financial assets, and risk factors.
The correct answer is 'True'.

Pareto efficiency is allocational efficiency, referring that in a financial equilibrium, allocations of assets are such that nobody's utility can decrease or increase by changing the allocations.
The correct answer is 'True'.

Hyperbolic discounting is time- consistent, classical discounting is not.
The correct answer is 'False'.

By definition, lottery A state dominates lottery B if A gives a strictly higher outcome than B in every state.
The correct answer is 'False',

If a utility function is strictly monotonic, then it is weakly monotonic as well.
The correct answer is 'True'.

If markets are complete, then all second period consumptions can be achieved by asset trades.
The correct answer is 'True'.

Risk neutral probabilities are representing the heterogeneous beliefs of the investors.
The correct answer is 'False'.

The efficient frontier is the set of all portfolios in the mean-variance opportunity set for which there is no other point in the mean-variance opportunity set that improves mean and variance.
The correct answer is 'True'.

The alpha is the difference between the mean of an asset and the risk-free return.
The correct answer is 'False'.

Combining two risky assets can yield a portfolio with zero variance if the two assets are perfectly positively correlated (correlation coefficient is +1).
The correct answer is 'False',

According to the Two-Fund Separation Theorem for mean-variance investors, all investors should invest to the market portfolio and to the risk free asset.
The correct answer is 'True'.

Financial distress: when a firm has trouble meeting its debt obligation.

Default: a firm that fails to make the required interest or principal payments on the debt.

- After the firm defaults, debt holders are given certain rights to the assets of the firm and may even take legal ownership of the firm's assets through bankruptcy.
- Equity financing does not carry this risk. Although equity holders hope to receive dividends, the firm is not legally obligated to pay them.

if the company fails ☹ economic distress

Company will go bankrupt. ☹ financial distress

Trade-off theory I. The firm picks its capital structure by trading off the benefits of the tax shield from debt against the costs of financial distress and agency costs.

The total value of a levered firm equals the value of the firm without leverage plus the present value of the tax savings from debt, less the present value of financial distress costs.

Why firms choose debt levels that are too low to fully exploit the interest tax shield.

Differences in the use of leverage across industries (due to differences in the magnitude of financial distress costs and the volatility of cash flows).

Agency costs: arise when there are conflicts of interest between the firm's stakeholders.

Agency benefit: Management entrenchment: facing little threat of being fired and replaced
☹ free to run the firm ☹ decision may benefit the managers more than the investors.

Asymmetric Information: A situation in which parties have different information.

Pecking order hypothesis: managers will prefer to fund investments by first using retained earnings, then debt, and equity only as a last resort.

Payout policy: the way a firm chooses between the alternative ways to distribute free cash flows to equity holders.

MM Dividend Irrelevance In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price.

Payout vs retention of cash: In perfect capital markets, once a firm has taken all positive-NPV investments, retention versus payout decision is irrelevant.

Modigliani-Miller Payout Irrelevance: In perfect capital markets, if a firm invests

excess cash flows in financial securities, the firm's choice of payout versus retention is irrelevant and does not affect the initial share price.

Spin-off: When a firm sells a subsidiary by selling shares in the subsidiary alone.

- Non-cash special dividends are commonly used to spin off assets or a subsidiary as a separate company.

Spin-Offs offer two advantages:

- It avoids the transaction costs associated with a subsidiary sale.
- The special dividend is not taxed as a cash distribution.

Stock dividend:

With a stock dividend, a firm does not pay out any cash to shareholders.

- ⌚ total market value of the firm's equity is unchanged.
- ⌚ The only thing that is different is the number of shares outstanding.
- ⌚ The stock price will therefore fall because the same total equity value is now divided over a larger number of shares.
- ⌚ Stock dividends are not taxed.

Stock splits:

- ⌚ The number of shares is proportionally increased, and the price per share is proportionally reduced so that there is no change in value.
- ⌚ The typical motivation for a stock split is to keep the share price in a range thought to be attractive to small investors.
- ⌚ On average, announcements of stock splits are associated with a 2% increase in the stock price.

Reverse split:

When the price of a company's stock falls too low and the company reduces the number of outstanding shares.

Dividend Smoothing: Firms change dividends infrequently, and dividends are much less volatile than earnings.

Dividend Signaling Hypothesis: The idea that dividend changes reflect managers' views about a firm's future earnings prospects ⌚ If firms smooth dividends, the firm's dividend choice will contain information regarding management's expectations of future earnings.

- When a firm increases its dividend, it sends a positive signal to investors that management expects to be able to afford the higher dividend for the foreseeable future.
- When a firm decreases its dividend, it may signal that management has given up hope that earnings will rebound in the near term and so need to reduce the dividend to save cash.

Working Capital Management

NWC is the capital required on the short term to run the business. Main components are cash, inventory, receivables, payables. Only that is needed to run the business on a day-to-day basis. It doesn't include excess cash!!!

Working capital ($\downarrow$) can alter a firm's value ($\uparrow$) by affecting its FCFF ($\uparrow$).

cash cycle: The length of time between when a firm pays cash to purchase its initial inventory and when it receives cash from the sale of the output produced from that inventory

Cash Conversion Cycle (CCC) = Accounts Receivable Days + Inventory Days – Accounts Payable Days

Accounts Receivable Days = $AR \times 365 / \text{sales}$

Inventory days = $\text{inv} \times 365 / (\text{Cost of sales} + \text{SG})$

Accounts Payable Days = $AP \times 365 / \text{cost of sales}$

Operation cycle= The average length of time between when a firm originally purchases its inventory and when it receives the cash back from selling its product.

Trade credit= 2/10 net 30

A firm should borrow using AP only if trade credit is the least expensive source of funding. ☐

The cost of the trade credit depends on the credit terms.

The higher the discount percentage offered, the greater the cost of forgoing the discount.

The shorter the loan period, the greater the cost of forgoing the discount.

Benefits of Holding Inventory:

Prevent stock-outs

Seasonality in demand

Costs of Holding Inventory:

Acquisition costs: are costs of the inventory itself over the period being analyzed.

Order costs: are the total costs of placing an order over the period being analyzed.

Carrying costs: storage cost, insurance, taxes, spoilage, obsolescence, opportunity costs of the funds tied up in inventory.

Just-in-Time (JIT) Inventory Management:

When a firm acquires inventory precisely when needed so that its inventory balance is always zero, or very close to it

JIT is often used to reduce carrying costs as much as possible

Matching Principle: a firm's short-term needs should be financed with short-term debt and long-term needs should be financed with long-term sources of funds.

Permanent Working Capital:

The amount that a firm must keep invested in its short-term assets to support its continuing operations ☐ the firm should finance this permanent investment in working capital with long-term sources of funds

Temporary Working Capital:

The difference between the actual level of short-term working capital needs and its permanent working capital requirements. → the firm should finance this temporary investment in working capital with short-term sources of funds.

Aggressive Financing Policy:

Financing part or all of a firm's permanent working capital with short-term debt
Risk of an aggressive financing policy is funding risk (the risk of incurring financial distress costs should a firm not be able to refinance its debt in timely manner or at a reasonable rate)

Conservative Financing Policy:

Financing all of a firm's permanent working capital with long-term debt

Raising Equity Capital

Angel investors / angel groups:

Individuals who buy equity in small private firms.

Venture capital firms:

A limited partnership that specializes in raising money to invest in the private equity of young firms

General partners are running the business → venture capitalist

Advantages to angels for limited partners:

More diversified portfolios

Benefit from the expertise of the GPs.

GPs charge large fees:

20% of any positive return

Annual management fee app. 2%

Private equity firms:

They finance mainly privately held firms rather than start-ups.

Leveraged buyout (LBO) → buying publicly held firms.

Institutional investors:

Pension funds, insurance companies, endowments, foundations (directly, or through VC-s/PE-s)

Corporate investors (= corporate partner, strategic partner, strategic investors)

Founding round = seed round

Each time the firm raises money → each set of securities with special terms and provisions

Preferred Stock

issued by mature companies usually has a preferential dividend and seniority in any liquidation and sometimes special voting rights.

issued by young companies has seniority in any liquidation but typically does not pay regular cash dividends and often contains a right to convert to common stock.

Convertible Preferred Stock

Preferred stock that gives the owner an option to convert it into common stock on some future date

Post-money value = Pre-money value + Amount invested

Percentage ownership = Amount invested / Post-money valuation

Exercise:

In addition to common shares, your firm raised \$10 million in Series A financing with a 2x liquidation preference, no participation rights, and a \$25 million post-money valuation.

What is the minimum sale price such that the Series A investors will convert their shares?

Series A investors have a liquidation preference of $2 \times \$10 \text{ million} = \20 million

Series A investors own 10/25 of the firm.

Because the Series A investors will receive up to 2x their investment from their liquidation preference, they will not be willing to convert their shares to common unless the value of the firm has at least doubled from \$25 million to \$50 million.

At \$50 million, Series A investors are willing to convert, and common shareholders will receive the remaining \$30 million.

Capital Structure I. –

Modigliani-Miller, Debt and taxes

Equity versus debt financing: Capital structure: The relative proportions of debt, equity, and other securities that a firm has outstanding.

Payments to debt holders are made before payments to equity holders!!!

Risk is greater for the levered equity → return should be greater, too!!!

Modigliani – Miller I: leverage, arbitrage and firm value perfect capital markets:

Investors and firms can trade the same set of securities at competitive market prices equal to the present value of their future cash flows.

There are no taxes, transaction costs, or issuance costs associated with security trading.

A firm's financing decisions do not change the cash flows generated by its investments, nor do they reveal new information about them

Proposition I: In a perfect capital market, the total value of a firm is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure.

Proposition II: The cost of capital of levered equity increases with the firm's market value debt-equity ratio

Homemade leverage: as long as investors can borrow or lend at the same interest rate as the firm, homemade leverage is a perfect substitute for the use of leverage by the firm → so why would someone give a higher value for a levered firm?

The value of a levered firm exceeds the value of the firm without leverage due to the PV of the tax savings from debt.

Mergers and Acquisitions

Why paying more than the current price?

Synergy: Two firms together add up for more: $2+2=5$, thus we pay a maximum of 3 for the firm.

Control premium: Should we control the firm, the value will climb from 2

Undervaluation „premium”: The fair firm value is 3 today, but the market price is only 2. We pay a maximum of 3, no need to own another firm..

The risk of synergies:

- 1) the transaction process jeopardises the customer connections.
- 2) competitors aim immediately at our best staff,
- 3) customers buying from both firms hit take some of their purchase to a competitor,
- 4) the remaining customers ask for a reduction as thanks to their loyalty

Costs of realization

Time need of realization

1. The fusion of similar-sized firms is less likely to be successful.
2. Synergies from immediate cost reductions (e.g. downsizing) are easier to capture than those linked to future growth, new products, or joint R&D.
3. Acquisitions focused on smaller private firms are more likely to be successful. The harder the competition, the more likely the loss in value is.
4. Hostile takeovers promise better management and results than friendly transactions.
5. Companies with outstanding efficiency are more likely to be able to create value by acquiring firms.
6. Firms carefully considering synergies

Multiple-based valuation methods

Comparable transactions- Based on earlier actual take-over transactions BUT The markets change over time Synergic premiums

Market multiples- Based on current stock exchange quotes BUT Market efficiency
Minority discount

What is the value of a book store with a sales of 200 M HUF if another book shop of 150 M HUF revenue has recently been sold for 160 M HUF?

$$P1/S1=160/150=1.07$$

$$P2=1.07*200=213.3 \text{ M HUF}$$

In multiples In valuation

P- Price of one single share

$P/E = \text{Price} / \text{EPS}$ $E / \text{Profit after tax}$
 $EV (\text{Enterprise value}) = E + D - \text{Cash}$
 $EV/BV (\text{Book Value}) = (V \pm \text{Cash}) / IC$
 $P/BV = E / \text{Book value of equity}$
 $P/S = \text{Price} / \text{Sales per share}$ E / Sales
 $EV/S = (V \pm \text{Cash}) / \text{Sales}$
 CF Operative CF (usually)

Grouping the multiples: Profit-based, Revenue-based; Asset-based; Industry specific

Choosing the right ratio:

Relevance: Either the numerator or the denominator should be some kind of value.

Consistency The numerator and the denominator should be of the same dimension (V or E).

Uniformity Same financial year, accounting standard, currency

The process of multiple-based valuation:

0. Pick a multiple.

1. Choose comparable firms. Similar risk, growth, profit generation

(No industry or size requirement theoretically.)

2. Control for differences. (Similar but not the same.)

Subjective corrections (weighted average)

Modified multiples (corrected for differences)

3. Check reasonability.

Industry regressions

Market regressions

Assumptions of multiple types 1

Profit-based (P/E, EV/EBIT): the same link between the right CF and the given profit.
(g, CF-generation, risk)

Revenue-based (EV/S): from a given amount of revenue any firm generates the same profit.
(g, CF-generation, risk, margin : CF/S)

Asset-based (P/BV, EV/IC): the assets generate the same profit everywhere.
(g, CF-generation, risk, ROI(c)/ROE)

Industry specific (EV/employee): the given specific variable creates the same profit everywhere
(g, CF-generation, risk, industry: same technology and efficiency)

Interpreting P/E 1

High P/E:
High dividend pay-off
Low $r-g$
 r big & g huge
 r small & g tiny
P/E/g (PEG)
Relative P/E = $(P/E_{\text{Firm}})/(P/E_{\text{Industry}})$

Revenue-based multiples Easy to use Always positive No accounting issues Very far from the CF

Asset-based multiples Stable, easy-to-compare, loss making firms create no issue
Accounting policy differences, off-balance sheet items, $BV(E) < 0$

Industry specific multiples

Value creation linked to operation No accounting issues No further information needed
Same technology and efficiency? Examples Number of subscribers, readers, visitors, buyers, patients, quantity produced, owned resource confirmed, sales area, number of rooms

Risk: more, but known possible future outcomes with given probabilities; the distribution is known.

- **Uncertainty:** more possible future outcome, but the probabilities or even all the possible outcomes are unknown.

Risk averse: prefers the certain option.

Risk neutral: indifferent between the two options.

Risk seeker: prefers the risky option.

Certainty equivalent

CE is the certain amount of money the decision maker would accept instead of the risky bet.

Risk averse: CE is lower than the expected value of the risky choice

Risk neutral: CE equals the expected value

Risk seeker: CE is higher than the expected value of the risky choice

Measuring risk

A risk measure characterizes the distribution of the risk factor (return of the investment).

- Standard deviation /variance
- Quantile (percentile)
- Maximum /minimum

Risk-return trade-off in investments

There is a trade off between risk and expected return

The higher the risk, the higher the expected return

Basel I. – credit risk

1996: Amendment – market risk of the trading book

2004: Basel II. – operational risk
2009: Basel II.5 – market risk revisited
Basel III. – liquidity risk, systemic risk

Capital Requirement Regulation (CRR) Calculation of own funds requirements:

- standardized approach (risk weights are set by the regulator)
- internal model based calculations

Var “What loss level is such that we are X% confident it will not be exceeded in N business days?”

Regulators have traditionally used VaR to calculate the capital they require banks to keep
The market-risk capital has been based on a 10-day VaR estimated where the confidence level is 99%

Credit risk and operational risk capital are based on a one-year 99.9% VaR

VaR is the loss level that will not be exceeded with a specified probability

Expected shortfall (ES) is the expected loss given that the loss is greater than the VaR level
(also called C-VaR and Tail Loss)

Regulators have indicated that they plan to move from using VaR to using ES for determining market risk capital

Two portfolios with the same VaR can have very different expected shortfalls

VaR satisfies the first three conditions but not the fourth one

ES satisfies all four conditions.

Spectral Risk Measures

1. A spectral risk measure assigns weights to quantiles of the loss distribution
2. VaR assigns all weight to Xth percentile of the loss distribution
3. Expected shortfall assigns equal weight to all percentiles greater than the Xth percentile
4. For a coherent risk measure weights must be a non-decreasing function of the percentiles

Bunching = Bunching occurs when exceptions are not evenly spread throughout the back testing period

1988: BIS Accord

Main Provisions:

- Capital must be 8% of risk weighted amount.
- At least 50% of capital must be Tier 1
- Tier 1 Capital: common equity, non-cumulative perpetual preferred shares
- Tier 2 Capital: cumulative preferred stock, certain types of 99-year debentures, subordinated debt with an original life of more than 5 years

Risk weighted amount (RWA) consists of

- Sum of products of risk weight times asset amount for on-balance sheet items
- Sum of products of risk weight times credit equivalent amount for derivatives and off-balance sheet commitments

Basel II Implemented in 2007 -- Three pillars

- New minimum capital requirements for credit and operational risk
- Supervisory review: more thorough and uniform
- Market discipline: more disclosure

Basel III

- Capital Definition and Requirements
- Capital Conservation Buffer
- Countercyclical Buffer
- Leverage Ratio
- Liquidity Ratios
- Capital for CVA Risk

Capital Definition and Requirements

- Three types:
 - Common equity Tier 1
 - Additional Tier 1
 - Tier 2
- Definitions tightened
- Limits
 - Common equity > 4.5% of RWA
 - Tier 1 > 6% of RWA
 - Tier 1 plus Tier 2 > 8% of RWA
- Capital levels implemented

Capital Conservation Buffer

- Extra 2.5% of common equity required in normal times to absorb losses in periods of stress
- If total common equity is less than 7% ($=4.5\%+2.5\%$) dividends are restricted

Countercyclical Buffer

- Extra equity capital to allow for cyclicalities of bank earnings
- Left to the discretion of national regulators
- Can be as high as 2.5% of RWA
- Dividends restricted when capital is below required level

D-SIBs are domestic systemically important banks.

G-SIBs are Global Systemically Important Banks

Estimating Default Probabilities

- Alternatives:
 - Use historical data
 - Use credit spreads
 - Use Merton's model Merton's model regards the equity as an option on the assets of the firm

● In a simple situation the equity value is $\max(VT - D, 0)$ where VT is the value of the firm and D is the debt repayment required

“Enterprise risk management is a process, effected by an entity’s board of directors, management, and other personnel, applied in strategy setting and across the enterprise, designed to identify potential events that may affect the entity, and manage risk to be within its risk appetite, to provide reasonable assurance regarding the achievement of entity objectives.”

A call is an option to buy

A put is an option to sell

A European option can be exercised only at the end of its life

An American option can be exercised at any time

Market Makers

Most exchanges use market makers to facilitate options trading

A market maker quotes both bid and ask prices when requested

The market maker does not know whether the individual requesting the quotes wants to buy or sell

Principal Protected Note

Allows investor to take a risky position without risking any principal

Example: \$1000 instrument consisting of

3-year zero-coupon bond with principal of \$1000

3-year at-the-money call option on a stock

portfolio currently worth \$1000

Box Spread

A combination of a bull call spread and a bear put spread

If all options are European a box spread is worth the present value of the difference between the strike prices

(CAPM) is a model that can be used to calculate the expected return from an asset during a period in terms of the risk of the return $ER_i = R_f + \beta_i(ER_m - R_f)$

Arbitrage pricing theory (APT) is a multi-factor asset pricing model based on the idea that an asset's returns can be predicted using the linear relationship between the asset’s expected return and a number of macroeconomic variables that capture systematic risk

$$E(R)_i = E(R)_z + (E(I) - E(R)_z) \times \beta_n$$

Unlike the CAPM, which assume markets are perfectly efficient, APT assumes markets sometimes misprice securities, before the market eventually corrects and securities move back to fair value. Using APT, arbitrageurs hope to take advantage of any deviations from fair market value.

